



Figure 2: All menus



Figure 3: Status > Traffic Accounting

The next important step is to upgrade from 2.0.3 to the latest release. Start by checking whether new updates are available—go to the **System > Update** page. Select **Refresh Update List** and check for the availability of new updates. Do apply all the available patches. After updates, IPCop will be upgraded to the latest version 2.0.6 (as of January 1, 2013).

A very helpful diagnostics screen, especially for flash installations, is the **Memory** section of the **Status > System status** page, which tells you whether sufficient memory is available. Flash-based IPCop installations stop functioning if the Ramdisk memory is full. IPCop supports alert emails; configure the required email settings to enable the feature.

One of the interesting features built into IPCop 2.0.x is traffic accounting, which monitors traffic volume. This requires no further explanation; just look at Figure 3.

Box 2: Stateful packet inspection

IPCop is a stateful packet inspection firewall—this means that all outgoing traffic from the Green network to the Red network is tracked. Only incoming traffic in response to outgoing requests is allowed inside. All other traffic from the Red network to Green (without a corresponding outgoing request) is logged and dropped.



Figure 4: Services > Proxy

The **Services > Proxy** menu is one of the most important configuration menus of IPCop. In this menu, configuring an upstream proxy allows IPCop to access the Internet via an external proxy server. This will be required for Internet connections requiring proxy connectivity or in a local environment. If required, the username and password for the proxy server can also be configured here.

IPCop can allow clients to access the Internet in the following two ways:

- **Transparent proxy:** This enables all the requests from the Web browser to be forwarded to the Internet, requiring no browser re-configuration. In this mode, URLFilter settings control the HTTP traffic. However, HTTPS traffic goes unfiltered. Also, there is a possibility of users bypassing the URL filter mechanism.
- **Non-transparent proxy:** This requires manual configuration of all browsers to use the Green IP address and proxy port of IPCop. Running in

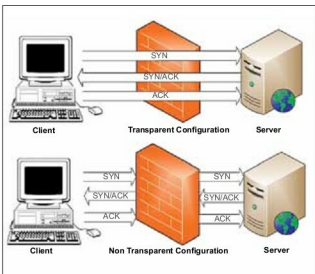


Diagram 3: Transparent and non-transparent proxies